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## Composition for regulating production of proteins

### Abstract

Some embodiments of the present disclosure relate to one or more compositions that upregulate the production of one or more sequences of mRNA. The sequences of mRNA may encode for translation of a target biomolecule, thereby causing an increase in the bioavailability of the target biomolecule within a subject that is administered the one or more compositions. In some embodiments of the present disclosure, the target biomolecule is a protein such as MANF.

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### Field of Classification Search

**CPC:** C12N (15/86); C12N (2750/14143); C07K (14/475)

### References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 11085055 | 12/2020 | Mallol et al. | N/A | N/A |
| 11162102 | 12/2020 | Minshull et al. | N/A | N/A |
| 11530423 | 12/2021 | Thompson | N/A | N/A |
| 11873505 | 12/2023 | Thompson | N/A | N/A |
| 12018274 | 12/2023 | Thompson | N/A | N/A |
| 12134770 | 12/2023 | Thompson | N/A | N/A |
| 2024/0026377 | 12/2023 | Thompson | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 2721333 | 12/2008 | CA | N/A |

**OTHER PUBLICATIONS**


O'Brien et al. “Overview of microRNA biogenesis, mechanisms of actions, and circulation.” Frontiers in endocrinology 9 (2018): 402. cited by applicant

Gorski et al. “RNA-based recognition and targeting: sowing the seeds of specificity.” Nature Reviews Molecular Cell Biology 18.4 (2017): 215-228. cited by applicant

Bottoni et al. “Targeting BTK through microRNA in chronic lymphocytic leukemia.” Blood, The Journal of the American Society of Hematology 128.26 (2016): 3101-3112. cited by applicant

Christensen et al. “Recombinant adeno-associated virus-mediated microRNA delivery into the postnatal mouse brain reveals a role for miR-134 in dendritogenesis in vivo.” Frontiers in neural circuits 3 (2010): 848. cited by applicant

Bofill-De Ros et al. “Guidelines for the optimal design of miRNA-based shRNAs.” Methods 103 (2016): 157-166. cited by applicant

Denzler et al. “Impact of microRNA levels, target-site complementarity, and cooperativity on competing endogenous RNA-regulated gene expression.” Molecular cell 64.3 (2016): 565-579. cited by applicant

Van Den Berg et al. “Design of effective primary microRNA mimics with different basal stem conformations.” Molecular Therapy Nucleic Acids 5 (2016). cited by applicant

Tritschler et al. “Concepts and limitations for learning developmental trajectories from single cell genomics.” Development 146.12 (2019): dev170506. cited by applicant

Ahmadzadeh et al. “BRAF mutation in hairy cell leukemia.” Oncology reviews 8.2 (2014): 253. cited by applicant

Patton et al. “Biogenesis, delivery, and function of extracellular RNA.” Journal of extracellular vesicles 4.1 (2015): 27494. cited by applicant

Clark et al. “Detection of BRAF splicing variants in plasma-derived cell-free nucleic acids and extracellular vesicles of melanoma patients failing targeted therapy therapies.” Oncotarget 11.44 (2020): 4016. cited by applicant

Kondratov et al. “Direct head-to-head evaluation of recombinant adeno-associated viral vectors manufactured in human versus insect cells.” Molecular Therapy 25.12 (2017): 2661-2675. cited by applicant

Nature (2010. Gene Expression. Scitable. Available online at Nature.com) <https://www.nature.com/scitable/topicpage/gene-expression-14121669> (2010). cited by applicant

Brutons Tyrosine Kinase Genbank Sequence (2023). cited by applicant

GenBank EGFR Sequence (2023). cited by applicant

GenBank EGF Sequence (2023). cited by applicant

NCBI search results for SEQ ID No. 5 (2024). cited by applicant

NCBI Nucleotide Sequence ALK Lingand, search performed Dec. 26, 2024 (2023). cited by

applicant
NCBI Nucleotide Sequence ALK Receptor, search performed Dec. 26, 2024 (2023). cited by applicant
NCBI Nucleotide Sequence for PARP, search performed Dec. 26, 2024 (2024). cited by applicant
GenBank FLT3 Sequence (2024). cited by applicant

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## Background/Summary

(1) This application contains a Sequence Listing electronically submitted via Patent Center to the United States Patent and Trademark Office as an XML Document file entitled “A8149889US-SequenceListing.xml” created on 2024 Oct. 23 and having a size of 16,186 bytes. The information contained in the Sequence Listing is incorporated by reference herein.
TECHNICAL FIELD
(2) The present disclosure generally relates to compositions for regulating the production of proteins. In particular, the present disclosure relates to compositions for regulating gene expression and, consequently, the production of proteins that act as neurotrophic factors.
BACKGROUND
(3) Aging in humans can result in decreased levels of neurotrophic factors.
(4) This can result in reduced tissue repair after tissue injury.
(5) As such, it may be desirable to establish therapies, treatments and/or interventions that may induce neurotrophic factors to be endogenously produced in a subject.
SUMMARY
(6) Some embodiments of the present disclosure relate to one or more compositions that upregulate the production of one or more sequences of messenger ribonucleic acid (mRNA). The sequences of mRNA may encode for the translation of a target biomolecule, thereby causing an increase in the bioavailability of the target biomolecule within a subject that is administered the one or more compositions. In some embodiments of the present disclosure, the target biomolecule is a protein such as mesencephalic astrocyte-derived neurotrophic factor (MANF).
(7) In some embodiments of the present disclosure the compositions comprise a plasmid of deoxyribonucleic acid (DNA) that includes one or more insert sequences of nucleic acids that encode for the production of mRNA and a backbone sequence of nucleic acids that facilitates the introduction of the one or more insert sequences into one or more of a subject's cells where it is thereby expressed and/or replicated. Expression of the one or more insert sequences by one or more cells of the subject results in an increased production of the mRNA and, consequently, increased translation of the target biomolecule by one or more of the subject's cells.
(8) Some embodiments of the present disclosure relate to a recombinant plasmid (RP). In some embodiments of the present disclosure, the RP comprises a nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 2. The nucleotide sequence of SEQ ID NO. 1 and SEQ ID NO. 2 encodes for one or more nucleotide sequences that encode for an mRNA sequence that encodes for the protein MANF.
(9) Some embodiments of the present disclosure relate to a method of making a composition/target cell complex. The method comprises a step of administering an RP comprising SEQ ID NO. 1 and

SEQ ID NO. 2 to a target cell for forming the composition/target cell complex, wherein the composition/target cell complex causes the target cell to increase the production of one or more sequences of mRNA that consequently increases the production of a target biomolecule.
(10) Embodiments of the present disclosure relate to at least one approach for inducing the endogenous production of one or more sequences of mRNA that encodes for a target biomolecule, for example MANF. A first approach utilizes gene vectors containing nucleotide sequences for increasing the endogenous production of one or more sequences of mRNA, which are complete or partial sequences of MANF and/or combinations thereof, which can be administered to a subject to increase the subject's production of one or more sequences of the mRNA.

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## Description

DETAILED DESCRIPTION
(1) Unless defined otherwise, all technical and scientific terms used therein have the meanings that would be commonly understood by one of skill in the art in the context of the present disclosure. Although any methods and materials similar or equivalent to those described therein can also be used in the practice or testing of the present disclosure, the preferred compositions, methods and materials are now described. All publications mentioned therein are incorporated therein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited.
(2) As used therein, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. For example, reference to “a composition” includes one or more compositions and reference to “a subject” or “the subject” includes one or more subjects.
(3) As used therein, the terms “about” or “approximately” refer to within about 25%, preferably within about 20%, preferably within about 15%, preferably within about 10%, preferably within about 5% of a given value or range. It is understood that such a variation is always included in any given value provided therein, whether or not it is specifically referred to.
(4) As used therein, the term “ameliorate” refers to improve and/or to make better and/or to make more satisfactory.
(5) As used therein, the term “cell” refers to a single cell as well as a plurality of cells or a population of the same cell type or different cell types. Administering a composition to a cell includes in vivo, in vitro and ex vivo administrations and/or combinations thereof.
(6) As used therein, the term “complex” refers to an association, either direct or indirect, between one or more particles of a composition and one or more target cells. This association results in a change in the metabolism of the target cell. As used therein, the phrase “change in metabolism” refers to an increase or a decrease in the one or more target cells' production of one or more proteins, and/or any post-translational modifications of one or more proteins.
(7) As used therein, the term “composition” refers to a substance that, when administered to a subject, causes one or more chemical reactions and/or one or more physical reactions and/or one or more physiological reactions and/or one or more immunological reactions in the subject. In some embodiments of the present disclosure, the composition is a plasmid vector.
(8) As used therein, the term “endogenous” refers to the production and/or modification of a molecule that originates within a subject.
(9) As used therein, the terms “production”, “producing” and “produce” refer to the synthesis and/or replication of DNA, the transcription of one or more sequences of RNA, the translation of one or more amino acid sequences, the post-translational modifications of an amino acid sequence, and/or the production of one or more regulatory molecules that can influence the production and/or functionality of an effector molecule or an effector cell. For clarity, “production” is also used therein to refer to the functionality of a regulatory molecule, unless the context reasonably indicates

otherwise.
(10) As used therein, the term “subject” refers to any therapeutic target that receives the composition. The subject can be a vertebrate, for example, a mammal including a human. The term “subject” does not denote a particular age or sex. The term “subject” also refers to one or more cells of an organism, an in vitro culture of one or more tissue types, an in vitro culture of one or more cell types, ex vivo preparations, and/or a sample of biological materials such as tissue, and/or biological fluids.
(11) As used therein, the term “target biomolecule” refers to a protein molecule that is found within a subject.
(12) As used therein, the term “target cell” refers to one or more cells and/or cell types that are affected, either directly or indirectly, by a biomolecule.
(13) As used therein, the term “therapeutically effective amount” refers to the amount of the composition used that is of sufficient quantity to ameliorate, treat and/or inhibit one or more of a disease, disorder or a symptom thereof. The “therapeutically effective amount” will vary depending on the composition used, the route of administration of the composition and the severity of the disease, disorder or symptom thereof. The subject's age, weight and genetic make-up may also influence the amount of the composition that will be a therapeutically effective amount.
(14) As used therein, the terms “treat”, “treatment” and “treating” refer to obtaining a desired pharmacologic and/or physiologic effect. The effect may be prophylactic in terms of completely or partially preventing an occurrence of a disease, disorder or symptom thereof and/or the effect may be therapeutic in providing a partial or complete amelioration or inhibition of a disease, disorder, or symptom thereof. Additionally, the term “treatment” refers to any treatment of a disease, disorder, or symptom thereof in a subject and includes: (a) preventing the disease from occurring in a subject which may be predisposed to the disease but has not yet been diagnosed as having it; (b) inhibiting the disease, i.e., arresting its development; and (c) ameliorating the disease.
(15) As used therein, the terms “unit dosage form” and “unit dose” refer to a physically discrete unit that is suitable as a unitary dose for patients. Each unit contains a predetermined quantity of the composition and optionally, one or more suitable pharmaceutically acceptable carriers, one or more excipients, one or more additional active ingredients, or combinations thereof. The amount of composition within each unit is a therapeutically effective amount.
(16) Where a range of values is provided therein, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limit of that range and any other stated or intervening value in that stated range, is encompassed within the disclosure. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also, encompassed within the disclosure, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the disclosure.
(17) In some embodiments of the present disclosure, the composition is a recombinant plasmid (RP) for introducing genetic material, such as one or more nucleotide sequences, into a target cell for reproduction or transcription of an insert that comprises one or more nucleotide sequences that are carried within the RP. In some embodiments of the present disclosure, the RP is delivered without a carrier, by a viral vector, by a protein coat, or by a lipid vesicle. In some embodiments of the present disclosure, the vector is an adeno-associated virus (AAV) vector.
(18) In some embodiments of the present disclosure, the insert comprises one or more nucleotide sequences that encode for the production of at least one sequence of mRNA that increases the production of target biomolecules, such as a protein.
(19) In some embodiments of the present disclosure, the target biomolecule is MANF.
(20) Some embodiments of the present disclosure relate to a composition that can be administered to a subject with a condition that results, directly or indirectly, from the dysregulated production of

a biomolecule. When a therapeutically effective amount of the composition is administered to the subject, the production and/or functionality of one or more of the subject's biomolecules may change as a result.

(21) In some embodiments of the present disclosure, the production and/or functionality of one or more of the subject's intermediary molecules may change in response to the subject receiving a therapeutic amount of the composition, thereby changing production of one or more DNA sequences, one or more RNA sequences, and/or one or more proteins that regulate the levels and/or functionality of the one or more intermediary molecules. The one or more intermediary molecules may regulate the subject's levels and/or functionality of the one or more biomolecules.

(22) In some embodiments of the present disclosure, administering a therapeutic amount of the composition to a subject upregulates the production, functionality or both of one or more sequences of mRNA that each encode for one or more biomolecules.

(23) In some embodiments of the present disclosure, the composition is an RP that may be used for gene therapy. The gene therapy is useful for increasing the subject's endogenous production of one or more sequences of mRNA that encode for a target biomolecule. For example, the RP can contain one or more nucleotide sequences that cause increased production of one or more nucleotide sequences that cause an increased production of one or more mRNA sequences that encode for a biomolecule, such as MANF.

(24) In some embodiments of the present disclosure, the delivery vehicle for the RP used for gene therapy may be a vector that comprises a virus that can be enveloped, or not (unenveloped), replication effective or not (replication ineffective), or combinations thereof. In some embodiments of the present disclosure, the vector is a virus that is not enveloped and not replication effective. In some embodiments of the present disclosure, the vector is a virus of the Parvoviridae family. In some embodiments of the present disclosure, the vector is a virus of the genus Dependoparvovirus. In some embodiments of the present disclosure, the vector is an adeno-associated virus (AAV). In some embodiments of the present disclosure, the vector is a recombinant AAV. In some embodiments of the present disclosure, the vector is a recombinant AAV6.2FF.

(25) In some embodiments of the present disclosure, the delivery vehicle for the RP used for gene therapy may be a protein coat.

(26) In some embodiments of the present disclosure, the delivery vehicle for the RP used for gene therapy may be a lipid vesicle.

(27) The embodiments of the present disclosure also relate to the administration of a therapeutically effective amount of the composition. In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to a patient is between about 10 and about 1×10.sup.16 TCID.sub.50/kg (50% tissue culture infective dose per kilogram of the patient's body mass). In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to the patient is about 1×10.sup.13 TCID.sub.50/kg. In some embodiments of the present disclosure, the therapeutically effective amount of the composition that is administered to a patient is measured in TPC/kg (total particle count of the composition per kilogram of the patient's body mass). In some embodiments of the present disclosure, the therapeutically effective amount of the composition is between about 10 and about 1×10.sup.16 TCP/kg.

(28) Some embodiments of the present disclosure relate to an adeno-associated virus (AAV) genome consisting of an RP that, when operable inside a target cell, will cause the target cell to produce an mRNA sequence that upregulates the production of a biomolecule, with an example being MANF. The RP is comprised of AAV2 inverted terminal repeats (ITRs), a composite CASI promoter, and a human growth hormone (HGH) signal peptide followed by a mRNA expression cassette encoding for MANF, followed by a Woodchuck Hepatitis Virus post-transcriptional regulatory element (WPRE) and a Simian virus 40 (SV40) polyadenylation (polyA) signal.

(29) TABLE-US-00001 SEQ ID NO. 1 (backbone sequence No. 1): 5′

TCTAGAGATAATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTCTT
AACTATGTTGCTCCTTTTACGCTATGTGGATACGCTGCTTTAATGCCTTTGTATCATG
CTATTGCTTCCCGTATGGCTTTCATTTTCTCCTCCTTGTATAAATCCTGGTTGCTGTCT
CTTTATGAGGAGTTGTGGCCCGTTGTCAGGCAACGTGGCGTGGTGTGCACTGTGTTT
GCTGACGCAACCCCCACTGGTTGGGGCATTGCCACCACCTGTCAGCTCCTTTCCGGG
ACTTTCGCTTTCCCCCTCCCTATTGCCACGGCGGAACTCATCGCCGCCTGCCTTGCCC
GCTGCTGGACAGGGGCTCGGCTGTTGGGCACTGACAATTCCGTGGTGTTGTCGGGGA
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TTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAATTTCACAAATAAAG
CATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCATCAATGTATCTTATCAT
GTCTGGATCTCGACCTCGACTAGAGCATGGCTACGTAGATAAGTAGCATGGCGGGTT
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TCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCC
GGGCGGCCTCAGTGAGCGAGCGAGCGCGCCAGCTGGCGTAATAGCGAAGAGGCCC
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CGATTGAGCGTCAAAATGTAGGTATTTCCATGAGCGTTTTTCCTGTTGCAATGGCTG
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CAGCAGAAGGACATTTTAGGACGGGACTTGGGTGACTCTAGGGCACTGGTTTTCTTT

CCAGAGAGCGGAACAGGCGAGGAAAAGTAGTCCCTTCTCGGCGATTCTGCGGAGGG ATCTCCGTGGGGCGGTGAACGCCGATGATGCCTCTACTAACCATGTTCATGTTTTCTT TTTTTTTCTACAGGTCCTGGGTGACGAACAGGGTACC 3′ SEQ ID NO. 2 (mRNA expression cassette No. 2 - MANF): 5′ GCCACCATGGCGACGGGTTCAAGAACTTCCCTACTTCTTGCATTTGGCCTGCTTTGTT TGCCGTGGTTACAGGAGGGCTCGGCAATGCGCCGCATGTGGGCGACCCAGGGCCTG GCGGTGGCGCTGGCGCTGAGCGTGCTGCCGGGCAGCCGCGCGCTGCGCCCGGGCGA TTGCGAAGTGTGCATTAGCTATCTGGGCCGCTTTTATCAGGATCTGAAAGATCGCGA TGTGACCTTTAGCCCGGCGACCATTGAAAACGAACTGATTAAATTTTGCCGCGAAGC GCGCGGCAAAGAAAACCGCCTGTGCTATTATATTGGCGCGACCGATGATGCGGCGA CCAAAATTATTAACGAAGTGAGCAAACCGCTGGCGCATCATATTCCGGTGGAAAAA ATTTGCGAAAAACTGAAAAAAAAAGATAGCCAGATTTGCGAACTGAAATATGATAA ACAGATTGATCTGAGCACCGTGGATCTGAAAAAACTGCGCGTGAAAGAACTGAAAA AAATTCTGGATGATTGGGGCGAAACCTGCAAAGGCTGCGCGGAAAAAAGCGATTAT ATTCGCAAAATTAACGAACTGATGCCGAAATATGCGCCGAAAGCGGCGAGCGCGCG CACCGATCTG 3′ SEQ ID NO. 3 = SEQ ID NO. 1 + SEQ ID NO. 2 5′ TCTAGAGATAATCAACCTCTGGATTACAAAATTTGTGAAAGATTGACTGGTATTCTT AACTATGTTGCTCCTTTTACGCTATGTGGATACGCTGCTTTAATGCCTTTGTATCATG CTATTGCTTCCCGTATGGCTTTCATTTTCTCCTCCTTGTATAAATCCTGGTTGCTGTCT CTTTATGAGGAGTTGTGGCCCGTTGTCAGGCAACGTGGCGTGGTGTGCACTGTGTTT GCTGACGCAACCCCCACTGGTTGGGGCATTGCCACCACCTGTCAGCTCCTTTCCGGG ACTTTCGCTTTCCCCCTCCCTATTGCCACGGCGGAACTCATCGCCGCCTGCCTTGCCC GCTGCTGGACAGGGGCTCGGCTGTTGGGCACTGACAATTCCGTGGTGTTGTCGGGGA AATCATCGTCCTTTCCTTGGCTGCTCGCCTGTGTTGCCACCTGGATTCTGCGCGGGAC GTCCTTCTGCTACGTCCCTTCGGCCCTCAATCCAGCGGACCTTCCTTCCCGCGGCCTG CTGCCGGCTCTGCGGCCTCTTCCGCGTCTTCGCCTTCGCCCTCAGACGAGTCGGATCT CCCTTTGGGCCGCCTCCCCGCCTAAGCTTATCGATACCGTCGAGATCTAACTTGTTTA TTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAATTTCACAAATAAAG CATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCATCAATGTATCTTATCAT GTCTGGATCTCGACCTCGACTAGAGCATGGCTACGTAGATAAGTAGCATGGCGGGTT AATCATTAACTACAAGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGC TCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCC GGGCGGCCTCAGTGAGCGAGCGAGCGCGCCAGCTGGCGTAATAGCGAAGAGGCCC GCACCGATCGCCCTTCCCAACAGTTGCGCAGCCTGAATGGCGAATGGAATTCCAGA CGATTGAGCGTCAAAATGTAGGTATTTCCATGAGCGTTTTTCCTGTTGCAATGGCTG GCGGTAATATTGTTCTGGATATTACCAGCAAGGCCGATAGTTTGAGTTCTTCTACTC AGGCAAGTGATGTTATTACTAATCAAAGAAGTATTGCGACAACGGTTAATTTGCGTG ATGGACAGACTCTTTTACTCGGTGGCCTCACTGATTATAAAAACACTTCTCAGGATT CTGGCGTACCGTTCCTGTCTAAAATCCCTTTAATCGGCCTCCTGTTTAGCTCCCGCTC TGATTCTAACGAGGAAAGCACGTTATACGTGCTCGTCAAAGCAACCATAGTACGCG CCCTGTAGCGGCGCATTAAGCGCGGCGGGTGTGGTGGTTACGCGCAGCGTGACCGC TACACTTGCCAGCGCCCTAGCGCCCGCTCCTTTCGCTTTCTTCCCTTCCTTTCTCGCC ACGTTCGCCGGCTTTCCCCGTCAAGCTCTAAATCGGGGGCTCCCTTTAGGGTTCCGA TTTAGTGCTTTACGGCACCTCGACCCCAAAAAACTTGATTAGGGTGATGGTTCACGT AGTGGGCCATCGCCCTGATAGACGGTTTTTCGCCCTTTGACGTTGGAGTCCACGTTCT TTAATAGTGGACTCTTGTTCCAAACTGGAACAACACTCAACCCTATCTCGGTCTATTC TTTTGATTTATAAGGGATTTTGCCGATTTCGGCCTATTGGTTAAAAAATGAGCTGATT TAACAAAAATTTAACGCGAATTTTAACAAAATATTAACGTTTACAATTTAAATATTT GCTTATACAATCTTCCTGTTTTTGGGGCTTTTCTGATTATCAACCGGGGTACATATGA TTGACATGCTAGTTTTACGATTACCGTTCATCGATTCTCTTGTTTGCTCCAGACTCTC

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GAAACCTGCAAAGGCTGCGCGGAAAAAAGCGATTATATTCGCAAAATTAACGAACT
GATGCCGAAATATGCGCCGAAAGCGGCGAGCGCGCACCGATCTG 3′

(30) As will be appreciated by those skilled in the art, because the recombinant plasmid is a circular vector, the one or more sequences of the mRNA expression cassettes may be connected at the 3′ end of SEQ ID NO. 1, as shown in SEQ ID NO. 3, or at the 5′ end of SEQ ID NO. 1.

(31) As will be appreciated by those skilled in the art, a perfect match of nucleotides with each of the mRNA expression cassette sequences is not necessary in order to have the desired result of increased bioavailability of the target biomolecule as a result of the target cell producing the mRNA sequence that codes for the expression of the target biomolecule. In some embodiments of the present disclosure, about 80% to about 100% nucleotide sequence matching with each of the mRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 85% to about 100% nucleotide sequence matching with each of the mRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 90% to about 100% nucleotide sequence matching with each of the mRNA expression cassettes causes the desired result. In some embodiments of the present disclosure, about 95% to about 100% nucleotide sequence matching with each of the mRNA expression cassettes causes the desired result.

Example 1—Expression Cassette

(32) Expression cassettes for expressing mRNA were synthesized. The synthesized mRNA expression cassettes were cloned into the pAVA-00200 plasmid backbone containing the CASI promoter, multiple cloning site (MCS), Woodchuck Hepatitis Virus post-transcriptional regulatory element (WPRE), and Simian virus 40 (SV40) polyadenylation (polyA) sequence, all flanked by

the AAV2 inverted terminal repeats (ITR). pAVA-00200 was cut with the restriction enzymes KpnI and XbaI in the MCS and separated on a 1% agarose gel. The band of interest was excised and purified using a gel extraction kit. Each mRNA expression cassette was amplified by polymerase chain reaction (PCR) using Taq polymerase and the PCR products were gel purified and the bands on interest were also excised and purified using a gel extraction kit. These PCR products contained the mRNA expression cassettes in addition to 15 base pair 5′ and 3′ overhangs that aligned with the ends of the linearized pAVA-00200 backbone. Using in-fusion cloning, the amplified mRNA expression cassettes were integrated with the pAVA-00200 backbone via homologous recombination. The resulting RP contained the following: 5′ ITR, CASI promoter, mRNA expression cassette, WPRE, SV40 polyA and ITR 3′.

## Claims

1. A composition that comprises a recombinant plasmid (RP) with a sequence of nucleotides that encodes a sequence of messenger ribonucleic acid (mRNA) that encodes for a protein, wherein the sequence of nucleotides is 95% to 100% identical to SEQ ID NO. 2.
2. The composition of claim 1, wherein the RP is configured to be delivered to a target cell.
3. The composition of claim 1, wherein the RP is encased in a protein coat, a lipid vesicle, or any combination thereof.
4. The composition of claim 1, wherein the RP is encased in a viral vector.
5. The composition of claim 4, wherein the viral vector is one of a double-stranded DNA virus, a single-stranded DNA virus, a single-stranded RNA virus, or a double-stranded RNA virus.
6. The compositions of claim 4, wherein the viral vector is an adeno-associated virus.
7. The composition of claim 1, wherein the protein is mesencephalic astrocyte-derived neurotrophic factor (MANF).
8. A composition that comprises a recombinant plasmid (RP) with a sequence of nucleotides for encoding a sequence of messenger ribonucleic acid (mRNA) that encodes for a protein, wherein the sequence of nucleotides is 95% to 100% identical to SEQ ID NO. 3, wherein the protein is mesencephalic astrocyte-derived neurotrophic factor (MANF).